SOCIOL 723: Social Statistics II

Week 2, Regression Inference and Robust Standard Errors

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Today

The Sampling Distribution of OLS

Hypothesis Testing

Heteroskedasticity and Robust Standard Errors

Clustered Data

The Bootstrap

What Robust Standard Errors Cannot Fix

★ = essential, you must understand this ★ = for understanding, not examined

Where We Left Off

Last week, for the bivariate model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \bar{X})\,\epsilon_i}{\sum_i (X_i - \bar{X})^2}, \qquad \operatorname{Var}(\hat{\beta}_1 \mid X) = \frac{\sum_i (X_i - \bar{X})^2 \operatorname{Var}(\epsilon_i \mid X)}{\left[\sum_i (X_i - \bar{X})^2\right]^2},$$

which collapses to $\sigma^2 / \sum_i (X_i - \bar{X})^2$ only if $\operatorname{Var}(\epsilon_i \mid X)$ is the same for every i.

Today's agenda

1. What is the sampling distribution of $\hat{\beta}_1$ as n grows?
2. How do we turn a standard error into a test and a confidence interval?
3. What if $\operatorname{Var}(\epsilon_i \mid X)$ is *not* constant? (Heteroskedasticity, then clustering.)
4. The bootstrap as a general-purpose alternative.
5. What robust standard errors can and cannot fix.

Reminder: the Two-Star Convention

Frames marked ★ are **essential**: problem sets and exams draw on them directly.
Frames marked ★ are here for *understanding*, not for evaluation: matrix derivations, asymptotic arguments, and the general formulas you will meet in the literature.

- Examinable this week: the scalar variance formula and where it comes from; what heteroskedasticity and clustering *are* and why they matter; how to read and choose a standard error; what a bootstrap does; and the list of problems robust standard errors do not solve.
- Every standard error in this course starts from the variance of a weighted sum of errors. HC estimators estimate the individual *variance* terms; cluster-robust estimators also estimate the within-cluster *covariance* terms. Hold on to that sentence.

Sampling Distribution

The Question, in Plain Words ★

Last week we computed $\hat{\beta}_1 = 0.119$: one more year of schooling goes with about 12% higher earnings in our sample of 3,509 people.

- The GSS could just as easily have interviewed a *different* 3,509 people. That sample would have produced a different number, perhaps 0.112, perhaps 0.127.
- So 0.119 is one draw from a lottery. Before treating it as evidence about the population, we need to know how much it moves from draw to draw.
- The standard error is exactly that: the give-or-take attached to the estimate. Formally, it is (an estimate of) the standard deviation of $\hat{\beta}_1$ across hypothetical repeated samples.

This week in one sentence

Everything called "statistical inference", standard errors, t-statistics, p-values, confidence intervals, is machinery for learning that give-or-take number from the single sample we actually have.

What a Sampling Distribution Is ★

$\hat{\beta}_1$ is a number computed from *your* sample. Had you drawn a different sample from the same population, you would have got a different number.

The thought experiment

Imagine drawing sample after sample from the population and computing $\hat{\beta}_1$ each time. The histogram of those numbers is the sampling distribution. Where it is centered tells you about bias; how wide it is is the true sampling standard deviation, and the *standard error* is our estimate of that width.

- You only ever see one draw. Everything in this lecture is about inferring the shape of that histogram from a single sample.
- In lab we run the experiment by simulation, which is the fastest way to make the idea concrete.

Two Theorems We Lean On All Semester ★

Both are statements about what happens to a *sample average* as n grows.

Law of large numbers (LLN)

The average of n independent draws settles down on the population mean: $\frac{1}{n}\sum_i Z_i \xrightarrow{p} E[Z]$ (read $\xrightarrow{p}$ as "settles down on"). Flip a fair coin 10 times and the share of heads could be anything; flip it 10,000 times and it sits within a whisker of 0.5.

Central limit theorem (CLT)

For an average of *independent* draws, the *shape* of the wobble around the truth becomes a bell curve, $\sqrt{n}\,(\bar{Z} - E[Z]) \xrightarrow{d} \mathcal{N}(0, \text{Var}(Z))$ (read $\xrightarrow{d}$ as "the distribution approaches"), whatever the distribution of Z itself, so long as its variance is finite.

Why we care: $\hat{\beta}_1$ is built from sample averages, so the LLN gives consistency and the CLT approximate normality. Thursday's lab watches the bell curve emerge.

The Sampling Distribution of $\hat{\beta}_1$, as n Grows

The big picture: keep drawing samples from the population; as n grows, we will show that $\hat{\beta}_1$'s sampling behaviour approximates a normal distribution, without ever assuming the errors are normal (A6):

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n}\sum_i (X_i - \bar{X})\epsilon_i}{\frac{1}{n}\sum_i (X_i - \bar{X})^2}.$$

A normal is pinned down by its *mean* and *variance*, so the work here is to find both, and let the CLT supply the *shape*.

The plan:

1. The denominator converges to a constant (LLN).
2. The numerator, magnified by $\sqrt{n}$, converges to a normal distribution with mean 0 (CLT) under strict exogeneity (A4: $E[\epsilon_i \mid X] = 0$).
3. A normal distribution divided by a settled constant is still normal.

Step 1: The Denominator Settles (LLN)

The denominator is the sample variance of X. By the law of large numbers

$$\bar{X} \xrightarrow{p} E[X] = \mu_X,$$
$$\frac{1}{n}\sum_i (X_i - \bar{X})^2 \xrightarrow{p} E[(X_i - \mu_X)^2] = \text{Var}(X_i)$$

- It settles on the population variance, a *positive constant.*
- As $n \to \infty$, the observed X's approach, in aggregate, the population distribution of X.
- From here on we may treat the denominator as the constant $\text{Var}(X_i)$. Every remaining question is about the numerator, whose terms we may likewise write with μ_X in place of $\bar{X}$: it is the average of the $(X_i - \mu_X)\,\epsilon_i$.

Step 2: The Numerator's Mean

The numerator averages the identically distributed terms $(X_i - \mu_X)\,\epsilon_i$, so its mean is the terms' common mean. Compute it:

$$\begin{aligned} E\big[(X_i - \mu_X)\,\epsilon_i\big] &= E\Big[\,E\big[(X_i - \mu_X)\,\epsilon_i \mid X_i\big]\Big] && \text{iterated expectations} \\ &= E\big[(X_i - \mu_X)\,E[\epsilon_i \mid X_i]\big] && \text{given } X_i,\ (X_i - \mu_X) \text{ is fixed} \\ &= E\big[(X_i - \mu_X)\cdot 0\big] \;=\; 0 && \text{A4: } E[\epsilon_i \mid X_i] = 0 \end{aligned}$$

- By the LLN the numerator settles on this mean, 0, so $\hat{\beta}_1 - \beta_1 \xrightarrow{p} 0/\operatorname{Var}(X_i) = 0$: consistency. As sample approaches population, $\hat{\beta}_1$ nails β_1.
- But a mean says nothing about *spread*, and spread is what a standard error measures. Next: the variance.

Step 3: The Numerator's Variance

Take $n = 3$: the numerator averages the three terms $(X_i - \mu_X)\,\epsilon_i$. The variance of a sum is every term's variance plus a covariance for every pair:

$$\mathrm{Var}(\text{numerator}) = \tfrac{1}{9}\Big[\underbrace{\mathrm{Var}\big((X_1-\mu_X)\epsilon_1\big) + \mathrm{Var}\big((X_2-\mu_X)\epsilon_2\big) + \mathrm{Var}\big((X_3-\mu_X)\epsilon_3\big)}_{\text{one variance per draw}}$$

$$+ \underbrace{2\,\mathrm{Cov}\big((X_1-\mu_X)\epsilon_1,\ (X_2-\mu_X)\epsilon_2\big) + \cdots}_{\text{one covariance per pair; 3 pairs}}\Big]$$

- A2 kills every covariance: draws are *independent*, so each Cov is 0. Clustering breaks exactly this step.
- Merging the remaining variance terms, with n draws, $\mathrm{Var}\big((X_i-\mu_X)\epsilon_i\big)/n$.
- The spread shrinks like $1/\sqrt{n}$: magnify by $\sqrt{n}$ to hold it steady. That is where the $\sqrt{n}$ comes from.

Step 4: The CLT Adds the Shape

The magnified numerator has mean 0 and variance $\operatorname{Var}\left((X_i - \mu_X)\epsilon_i\right)$; the CLT adds the missing *shape*. Divide by the settled denominator:

$$\sqrt{n}\,(\hat{\beta}_1 - \beta_1) = \frac{\overbrace{\sqrt{n}\cdot\frac{1}{n}\sum_i (X_i - \bar{X})\,\epsilon_i}^{\xrightarrow{d}\ \mathcal{N}(0,\ \operatorname{Var}((X_i-\mu_X)\epsilon_i))}}{\underbrace{\frac{1}{n}\sum_i (X_i - \bar{X})^2}_{\xrightarrow{p}\ \operatorname{Var}(X_i)}} \xrightarrow{d} \mathcal{N}(0,\ V), \quad V = \frac{\operatorname{Var}\left((X_i - \mu_X)\epsilon_i\right)}{\left[\operatorname{Var}(X_i)\right]^2}.$$

- Under A5, homoskedasticity, $\operatorname{Var}(\epsilon_i \mid X_i) = \sigma^2$: the error variance is the same number at every value of X. The numerator of V then collapses and $V = \sigma^2/\operatorname{Var}(X_i)$, the familiar formula: next frame, slowly.
- Without A5, estimating V is this lecture's entire subject.

Under A4 and A5, the Meat Collapses

The meat is a mean square around the unconditional mean, so A4 first certifies that mean is zero; then A5 collapses it:

$$
\begin{aligned}
E\big[(X_i - \mu_X)^2\,\epsilon_i^2\big] &= E\Big[\,E\big[(X_i - \mu_X)^2\,\epsilon_i^2 \mid X_i\big]\,\Big] && \text{iterated expectations} \\
&= E\big[(X_i - \mu_X)^2\,E[\epsilon_i^2 \mid X_i]\big] && \text{given } X_i,\ (X_i - \mu_X)^2 \text{ is fixed} \\
&= E\big[(X_i - \mu_X)^2\,\sigma^2\big] && \text{A4 + A5: } E[\epsilon_i^2 \mid X_i] = \sigma^2 \\
&= \sigma^2\,E\big[(X_i - \mu_X)^2\big] = \sigma^2\ \mathrm{Var}(X_i) && \sigma^2 \text{ is one number: pull it out}
\end{aligned}
$$

So $V = \sigma^2\,\mathrm{Var}(X_i)/[\mathrm{Var}(X_i)]^2 = \sigma^2/\,\mathrm{Var}(X_i)$: the same answer as Week 1.

Testing

The Vocabulary, Pinned Down ★

You met these words in *Social Statistics I*; now they have foundations.

- A **null hypothesis** H_0 is a specific claim about a population quantity, e.g. $\beta_1 = 0$: schooling and earnings are unrelated.
- A **test statistic** measures the distance between estimate and null *in standard-error units*: $t = (\hat{\beta}_1 - 0)/\text{se}(\hat{\beta}_1)$. Big distance, implausible null.
- The *p*-**value**: *if* H_0 were true, how often would sampling noise alone produce a t this large? It is a statement about the data under the null, not the probability the null is true.
- A **95% confidence interval** collects every null value the data cannot reject; across repeated samples, intervals built this way cover the truth 95% of the time. *This* interval either does or does not; the 95% describes the procedure.

Testing a Single Coefficient ★

To test $H_0 : \beta_j = \beta_j^0$, compare

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\mathrm{se}(\hat{\beta}_j)}$$

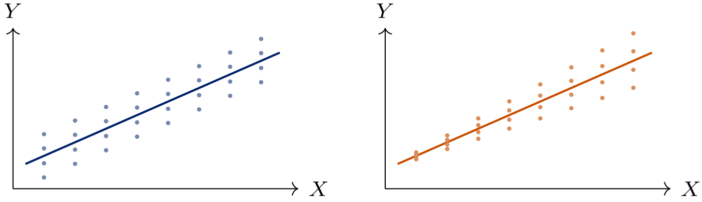
against t_{n-k} critical values; a 95% confidence interval is $\hat{\beta}_j \pm t_{0.975,\,n-k}\,\mathrm{se}(\hat{\beta}_j)$.

- By the last section, t is approximately $\mathcal{N}(0, 1)$ for large n, and t_{n-k} merges into that same curve (the critical values differ by under 2% once $n - k > 100$); under the classical assumption of normal errors, t_{n-k} is even exact at any n, which is where the convention comes from. This is why regression output reports t_{n-k}; clustered data will later change the right degrees of freedom.
- Everything depends on whether se is right. A t of 3.0 computed from a standard error that is half of what it should be is really a t of 1.5.

Heteroskedasticity

What Heteroskedasticity Is ★

Week 1's A5 had *two* parts: constant variance, and zero covariance across observations. This section drops the first while keeping the second; the clustering section then drops the second. **Homoskedasticity** says $\text{Var}(\epsilon_i \mid X_i) = \sigma^2$: the spread of the outcome around the line is the same everywhere.



homoskedastic heteroskedastic

Under exogeneity, both lines are estimated without bias: heteroskedasticity by itself does not bias $\hat{\beta}$. Only the *uncertainty* about them differs, and only the right-hand panel needs a different formula.

Heteroskedasticity Is the Norm

- Earnings dispersion rises steeply with education and experience: at 8 years of schooling almost everyone earns a modest wage; at 20 years the range is enormous.
- Group-level outcomes built from different numbers of individuals, a county mean over 50 people is noisier than one over 50,000.
- Binary outcomes: $\mathrm{Var}(Y_i \mid X_i) = p(X_i)(1 - p(X_i))$ is *necessarily* heteroskedastic, so the linear probability model always violates A5.

What it does and does not do

Does not: bias $\hat{\beta}$ or break consistency (given A4 and random sampling).
Does: invalidate the standard errors your software prints by default, so t-statistics, p-values and confidence intervals are wrong.

The Scalar Sandwich ★

The last section already handed us the bell's variance, built from population moments:

$$V = \frac{E\left[(X_i - \mu_X)^2\, \epsilon_i^2\right]}{\left[\,\mathrm{Var}(X_i)\right]^2} = \frac{\text{“meat”}}{\text{“bread”}^2}.$$

- The meat is a $(X_i - \mu_X)^2$-weighted average of the squared errors: errors sitting far from the mean of X count the most.
- Under A5 the squared error averages to σ^2 at every value of X, so the meat factors into $\sigma^2\ \mathrm{Var}(X_i)$ and $V = \sigma^2 / \mathrm{Var}(X_i)$: the classical formula is the A5 special case.
- Without A5 nothing factors, and the classical formula is simply wrong. But every piece of V is a population mean, and sample averages estimate population means. Next frame.

White's Estimator, in Scalar Form ★

Estimate each population mean by its sample counterpart, with residuals standing in for the unobservable errors: writing $d_i \equiv X_i - \bar{X}$, the bread $\mathrm{Var}(X_i)$ becomes $\frac{1}{n}\sum_i d_i^2$ and the meat $E[(X_i - \mu_X)^2 \epsilon_i^2]$ becomes $\frac{1}{n}\sum_i d_i^2\, \hat{e}_i^2$. The squared standard error of $\hat{\beta}_1$ is this estimate of V, divided by n; the $1/n$'s cancel into

$$\boxed{\hat{V}_{\mathrm{HC0}}(\hat{\beta}_1) = \frac{\sum_i d_i^2\, \hat{e}_i^2}{\left(\sum_i d_i^2\right)^2}} \qquad \text{versus} \qquad \hat{V}_{\mathrm{classical}}(\hat{\beta}_1) = \frac{s^2}{\sum_i d_i^2}.$$

- That is the whole of "heteroskedasticity-robust" and "(Eicker–)Huber–White" (White 1980); `Stata`'s `robust` adds HC1 scaling, next frame.
- With controls, replace d_i by Week 1's FWL residual $\tilde{d}_i$: $\hat{V}(\hat{\beta}_j) = \sum_i \tilde{d}_i^2 \hat{e}_i^2 / \left(\sum_i \tilde{d}_i^2\right)^2$: same formula, residualized weights.
- The classical formula pools *one* s^2; the robust one lets each observation carry its own $\hat{e}_i^2$.

The General Statement ★

For the full coefficient vector, with X_i now the *vector* of observation i's regressors and sample averages written as $\mathbb{E}_n$:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \left(\mathbb{E}_n[X_i X_i']\right)^{-1} \mathbb{E}_n[X_i \epsilon_i], \qquad \mathbb{E}_n[X_i X_i'] = \frac{1}{n} \sum_i X_i X_i' = \frac{1}{n} \boldsymbol{X}'\boldsymbol{X},$$

so this is Thursday's $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\epsilon}$, each factor scaled by $\frac{1}{n}$. The LLN gives $\mathbb{E}_n[X_i X_i'] \xrightarrow{p} E[X_i X_i'] \equiv \boldsymbol{Q}$ and $\mathbb{E}_n[X_i \epsilon_i] \xrightarrow{p} \mathbf{0}$, so $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}$ (*consistency*). Multiplying by $\sqrt{n}$ and applying the CLT to the second factor,

$$\boxed{\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(\mathbf{0},\ \boldsymbol{Q}^{-1}\boldsymbol{\Omega}\boldsymbol{Q}^{-1}\right)}, \qquad \boldsymbol{\Omega} = E[\epsilon_i^2 X_i X_i'].$$

Note what is *not* assumed: no normality, no homoskedasticity.

Reading the "Sandwich" ★

$$\boldsymbol{V} = \underbrace{\boldsymbol{Q}^{-1}}_{\text{bread}} \underbrace{\boldsymbol{\Omega}}_{\text{meat}} \underbrace{\boldsymbol{Q}^{-1}}_{\text{bread}}, \qquad \boldsymbol{Q} = E[X_i X_i'], \quad \boldsymbol{\Omega} = E[\epsilon_i^2 X_i X_i'].$$

Estimated by sample analogues, the $1/n$'s cancel into Thursday's matrices:

$$\widehat{\mathrm{Var}}_{\mathrm{HC0}}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\Big(\sum_i \hat{e}_i^2\, X_i X_i'\Big)(\boldsymbol{X}'\boldsymbol{X})^{-1}:$$

what `vcovHC` computes; the clustered version later swaps only the middle sum.

- The **bread** depends only on the regressors; the **meat** is where every assumption about the errors lives. Classical, robust, clustered, and HAC differ only in the meat.
- If $E[\epsilon_i^2 \mid X_i] = \sigma^2$, the meat is $\sigma^2\boldsymbol{Q}$ and $\boldsymbol{V} = \sigma^2\boldsymbol{Q}^{-1}$: the classical formula.

The organizing idea for the estimator

Keep the estimator $\hat{\beta}$ and the bread; change only the meat.

From Scalar to Matrix: the Dictionary ★

One change of meaning: X_i now denotes the *vector* of observation i's regressors, in the bivariate case $X_i = (1,\ X_i)'$. Then

$$X_i X_i' = \begin{pmatrix} 1 & X_i \\ X_i & X_i^2 \end{pmatrix}, \qquad X_i \epsilon_i = \begin{pmatrix} \epsilon_i \\ X_i \epsilon_i \end{pmatrix},$$

and their sample averages stack the scalar sums you have used all lecture:

$$\mathbb{E}_n[X_i X_i'] = \begin{pmatrix} 1 & \bar{X} \\ \bar{X} & \frac{1}{n}\sum_i X_i^2 \end{pmatrix}, \qquad \mathbb{E}_n[X_i \epsilon_i] = \begin{pmatrix} \bar{\epsilon} \\ \frac{1}{n}\sum_i X_i \epsilon_i \end{pmatrix}.$$

- Population versions: $\boldsymbol{Q} = E[X_i X_i']$ and the meat $\boldsymbol{\Omega} = E[\epsilon_i^2 X_i X_i']$, which stacks $E[\epsilon_i^2]$, $E[X_i \epsilon_i^2]$, $E[X_i^2 \epsilon_i^2]$.
- These are the objects inside the sandwich you just saw. The next frames multiply it out until the slope's entry is *exactly* the scalar V.

The Dictionary, Cashed: Multiply the Sandwich Out ★

Every symbol below is a population *number*. Abbreviate $v = \mathrm{Var}(X_i)$, $q = E[X_i^2]$, and the meat's three entries $a = E[\epsilon_i^2]$, $b = E[X_i \epsilon_i^2]$, $c = E[X_i^2 \epsilon_i^2]$. The ingredients ($\boldsymbol{Q}^{-1}$ by the 2×2 inverse, $\det \boldsymbol{Q} = q - \mu_X^2 = v$):

$$\boldsymbol{Q}^{-1} = \frac{1}{v} \begin{pmatrix} q & -\mu_X \\ -\mu_X & 1 \end{pmatrix}, \qquad \boldsymbol{\Omega} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

First multiplication, plain row-times-column arithmetic:

$$\boldsymbol{\Omega} \boldsymbol{Q}^{-1} = \frac{1}{v} \begin{pmatrix} aq - b\mu_X & b - a\mu_X \\ bq - c\mu_X & c - b\mu_X \end{pmatrix}.$$

One more multiplication by $\boldsymbol{Q}^{-1}$ from the left: next frame.

The Final 2×2 Matrix ★

Multiply again, row times column; for instance the bottom-right entry is $(-\mu_X)(b - a\mu_X) + 1 \cdot (c - b\mu_X) = a\mu_X^2 - 2b\mu_X + c$. Collecting all four entries:

$$\boldsymbol{Q}^{-1}\boldsymbol{\Omega}\boldsymbol{Q}^{-1} = \frac{1}{v^2}\begin{pmatrix} aq^2 - 2bq\mu_X + c\mu_X^2 & b(q + \mu_X^2) - \mu_X(aq + c) \\ b(q + \mu_X^2) - \mu_X(aq + c) & a\mu_X^2 - 2b\mu_X + c \end{pmatrix}.$$

The highlighted slope entry, with a, b, c substituted back:

$$a\mu_X^2 - 2b\mu_X + c = E[\mu_X^2\epsilon_i^2 - 2\mu_X X_i\epsilon_i^2 + X_i^2\epsilon_i^2] = E[(X_i - \mu_X)^2\epsilon_i^2],$$

so the $(2, 2)$ entry is $E[(X_i - \mu_X)^2\epsilon_i^2]/v^2 = V$: the scalar sandwich, recovered.

Reading the rest: the diagonal holds the intercept's and the slope's variances, the off-diagonal their covariance.

Why the Squared Residuals Are Systematically Too Small

The meat is estimated by an average, and averages forgive noise: each $d_i^2 \hat{e}_i^2$ may be a poor estimate of its own term, yet the mean still finds the target. What an average does not forgive is an error shared by every term – and the squared residuals carry one.

- OLS chooses the line that makes the residuals as small as possible, so it partly fits the noise itself: residuals run systematically *smaller* than the errors they stand in for.
- The exact bookkeeping (under A5, for clean arithmetic): $E[\hat{e}_i^2 \mid X] = \sigma^2(1 - h_{ii})$, where the leverage $h_{ii} = \frac{1}{n} + d_i^2 / \sum_j d_j^2$ measures how unusual X_i is.
- So every $\hat{e}_i^2$ is deflated, and most at *high-leverage* observations, exactly the ones the meat weights most heavily. The meat estimate, hence HC0, is biased down in small samples.
- As n grows, $h_{ii} \to 0$ and the deflation vanishes: a purely finite-sample issue. The target is still only the meat; the fixes just undo a known bias in its ingredients.

Where $\sigma^2(1 - h_{ii})$ Comes From ★

Write the residual in terms of the errors, step by step:

$$\begin{aligned}
\hat{e}_i &= Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i && \text{definition} \\
&= \epsilon_i - \bar{\epsilon} - (\hat{\beta}_1 - \beta_1)\, d_i && \text{sub } Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,\ \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \\
&= \epsilon_i - \bar{\epsilon} - d_i \frac{\sum_j d_j \epsilon_j}{\sum_j d_j^2} = \epsilon_i - \sum_j w_{ij}\, \epsilon_j && \text{Week 1: } \hat{\beta}_1 - \beta_1 = \textstyle\sum_j d_j \epsilon_j / \sum_j d_j^2
\end{aligned}$$

with $w_{ij} = \frac{1}{n} + d_i d_j / \sum_k d_k^2$: the fitted line absorbs a weighted share of *every* error, with weight $w_{ii} = h_{ii}$ on observation i's own. Under A5 and uncorrelated errors,

$$E[\hat{e}_i^2 \mid X] = \sigma^2 \Big(1 - 2h_{ii} + \sum_j w_{ij}^2\Big) = \sigma^2\,(1 - h_{ii}),$$

because $\sum_j w_{ij}^2 = h_{ii}$ (expand the square; $\sum_j d_j = 0$).

Reading: observation i pulls the line toward itself with strength h_{ii}; the residual keeps the $(1 - h_{ii})$ share. At $n = 2$, $h_{ii} = 1$: the line passes through both points and both residuals are exactly zero.

Finite-Sample Corrections: HC0–HC3 ★

Every $\hat{e}_i^2$ is deflated by its leverage. The corrections rescale each squared residual before it enters the meat, undoing the deflation (factors motivated by the homoskedastic benchmark, used regardless):

	Meat uses	Comment
HC0	$\hat{e}_i^2$	White's original; anticonservative when n is small
HC1	$\frac{n}{n-k}\hat{e}_i^2$	simple d.f. correction; `Stata`'s `robust` default
HC2	$\dfrac{\hat{e}_i^2}{1-h_{ii}}$	exact at the homoskedastic benchmark
HC3	$\dfrac{\hat{e}_i^2}{(1-h_{ii})^2}$	jackknife-like; recommended default

All four are consistent under any heteroskedasticity pattern and coincide as $n \to \infty$. MacKinnon and White (1985), Long and Ervin (2000): with $n < 250$ or high leverage, use **HC3**.

How Much Does It Matter?

- Robust standard errors can be *larger or smaller* than classical ones. There is no general direction.
- The difference is driven by whether the error variance happens to be large where X is extreme. In our GSS earnings regression it is modest: $\text{se}(\hat{\beta}_{\texttt{educ}})$ goes from 0.00510 (classical) to 0.00547 (HC0) to 0.00549 (HC3), about 7%.
- With a skewed or sparse regressor, a rare binary treatment, a long-tailed count; the difference can be a factor of two.
- Clustering, by contrast, routinely changes standard errors by factors of three to ten. In most sociological data that is the bigger issue.

Should You *Test* for Heteroskedasticity?

Classical tests exist: Breusch–Pagan, White.

Recommendation: no

Do not pre-test and then choose. Just report robust standard errors.

- A pre-test makes your final inference conditional on the pre-test outcome, which distorts the very sampling distribution you are claiming.
- The cost of robust standard errors when the errors happen to be homoskedastic is small, a modest efficiency loss in the *variance estimate*.
- The cost of classical standard errors when they are wrong is invalid inference.
- Exception: the tests are useful *diagnostically*. Heteroskedasticity is often a symptom of a misspecified functional form, which is worth knowing about for its own sake.

Clustering

Why Independence Fails ★

Sociological data are almost never i.i.d. draws. Three familiar designs, and what goes wrong:

- **Students within schools; workers within firms; siblings within families.** Two students in one school share teachers, curriculum, and neighborhood. If one does surprisingly well, the other probably does too: *knowing one error is informative about the other*. That is dependence.
- **Multi-stage surveys (the GSS).** The survey first draws *tracts*, then people within them. Which tracts got drawn is luck, and everyone from a drawn tract carries that tract's peculiarities into the sample *together*.
- **Treatment assigned to groups.** A state law hits everyone in the state at once, and same-state people already share a lot (economy, institutions, history): *the treated arrive in bundles of similar people*, not as independent draws.

The common thread: within a group, observations share something – a shock, a sampling draw, a treatment draw – and shared is the opposite of independent.

The Error Components Model ★

Formalize the shared piece:

$$Y_{ig} = \beta_0 + \beta_1 X_{ig} + \epsilon_{ig}, \qquad \epsilon_{ig} = \underbrace{\nu_g}_{\text{one draw per cluster}} + \underbrace{\eta_{ig}}_{\text{one draw per person}}$$

Each person has a true individual-level idiosyncrasy η_{ig}, independent across units and independent of the group shock; everyone in cluster g shares the one group-level shock ν_g. A shared component is a covariance:

$$\mathrm{Cov}(\epsilon_{ig}, \epsilon_{jg}) = \mathrm{Cov}(\nu_g + \eta_{ig},\ \nu_g + \eta_{jg}) = \mathrm{Var}(\nu_g) = \sigma_\nu^2 \qquad (i \neq j).$$

Different clusters share nothing (different ν's and η's): across clusters, independence still holds.

With $\mathrm{Var}(\epsilon_{ig}) = \sigma_\nu^2 + \sigma_\eta^2$, any two same-cluster errors have correlation $\rho_e = \sigma_\nu^2/(\sigma_\nu^2 + \sigma_\eta^2)$: the intra-class correlation, the share of error variance that is shared.

Where the Scalar Formula Breaks

Run Step 3's expansion once more, unconditionally, on the mean-zero terms $(X_i - \mu_X)\,\epsilon_i$. What changed is A2: clusters are independent draws; people within a cluster are not.

$$\mathrm{Var}\left(\tfrac{1}{n}\textstyle\sum_i (X_i - \mu_X)\epsilon_i\right) = \tfrac{1}{n^2}\Bigg[\underbrace{\sum_i \mathrm{Var}\left((X_i - \mu_X)\epsilon_i\right)}_{\text{what White's meat estimates}}$$

$$+ \underbrace{\sum_{\substack{i\neq j \\ \text{same cluster}}} \mathrm{Cov}\left((X_i - \mu_X)\epsilon_i,\ (X_j - \mu_X)\epsilon_j\right)}_{\text{assumed zero so far}} \Bigg]$$

Cross-cluster pairs drop out; survivors ride the shared shock ν_g (last frame).

- **Sign.** With positively correlated errors and same-sign $(X_i - \mu_X)$ within a cluster (a cluster-level treatment guarantees this), every surviving term is positive: the truth is larger than our formula.
- **Count.** A cluster of size m contributes $m(m-1)$ surviving pairs. Even a tiny correlation, multiplied by that many terms, can dominate.

The Cluster-Robust Fix, in Scalar Form

Estimate each cluster's mean square by its sample counterpart, exactly White's move: residuals for the unobservable errors, $\bar{X}$ for μ_X,

$$S_g = \sum_{i \in g} d_i \, \hat{e}_i \qquad \text{(one number per cluster)},$$

and place the summed squares over the bread:

$$\boxed{\hat{V}_{\text{CR}}(\hat{\beta}_1) = \frac{\sum_{g=1}^{G} S_g^2}{\left(\sum_i d_i^2\right)^2}}$$

Sum within clusters first, then square. The order of those two operations is the entire difference from White's estimator, which squares first and sums after.

Why Summing First Works

Expand one cluster's squared total:

$$S_g^2 = \Big(\sum_{i \in g} d_i \hat{e}_i\Big)^2 = \underbrace{\sum_{i \in g} d_i^2 \hat{e}_i^2}_{\text{White's terms}} + \underbrace{\sum_{i \neq j \in g} d_i d_j \, \hat{e}_i \hat{e}_j}_{\text{the cross terms, now estimated}} .$$

- Squaring *after* summing keeps every within-cluster product $\hat{e}_i \hat{e}_j$: exactly the terms the last frames showed we were wrongly assuming away. Across clusters, products never form: still assumed zero, which is the design claim that clusters are independent.
- With one observation per cluster, $S_g^2 = d_i^2 \hat{e}_i^2$ and this collapses exactly to HC0: clustering strictly generalizes White.

Cluster-Robust Variance (Liang–Zeger) ★

Let $g = 1, \ldots, G$ index clusters, with $\boldsymbol{X}_g$ and $\hat{\boldsymbol{e}}_g$ the stacked regressors and residuals in cluster g:

$$\boxed{\hat{V}_{\text{CR}}(\hat{\boldsymbol{\beta}}) = c\,(\boldsymbol{X}'\boldsymbol{X})^{-1}\Big(\sum_{g=1}^{G} \boldsymbol{X}_g'\hat{\boldsymbol{e}}_g\hat{\boldsymbol{e}}_g'\boldsymbol{X}_g\Big)(\boldsymbol{X}'\boldsymbol{X})^{-1}}$$

with the usual small-sample scaling $c = \frac{G}{G-1} \cdot \frac{n-1}{n-k}$ (CR1; `Stata`'s default).

- Same sandwich; the meat now sums outer products of *cluster* score vectors instead of individual ones, which is exactly how the cross terms get included.
- Allows arbitrary correlation within cluster (including serial correlation over time) and any heteroskedasticity, while assuming independence *across* clusters.
- Consistency is in $G \to \infty$, not $n \to \infty$.

At What Level Should You Cluster? ★

Abadie et al. (2023): clustering is a statement about the *design*, not a knob to tune.

- **Sampling-based**: cluster at the level of the sampling units, when the clusters in your data are a small share of the clusters in the population.
- **Design-based**: cluster at the level at which *treatment* is assigned. If a policy varies by state, cluster on state, even with individual-level data.
- Cluster at the level the design justifies; the right level, not the biggest.
- Never choose the level by whichever gives the smallest p-value.

Which Level? Three Examples, and Sampling Designs ★

- **Treatment at state, clustered by county: wrong.** Everyone in a state shares the state shock, so errors correlate *across counties* within a state; county-level clustering leaves those covariances out, and the SE is still too small. Too fine never works.
- **Treatment at county, clustered by state: allowed, not free.** Coarser is a *more generous assumption*: it allows errors to correlate across counties within a state instead of assuming they do not. The price is far fewer clusters, hence a noisier variance estimate (unreliable, *not* systematically larger); if the extra correlation is not there, you paid for nothing.
- **Treatment at state, data sampled in clusters (the GSS).** Now two lotteries create dependence: the *sampling* lottery (which tracts got drawn) and the *assignment* lottery (which states got the policy). Tracts are nested inside states, so same-tract pairs are also same-state pairs, and clustering by state covers both at once. With nested lotteries, cluster at the coarsest one.

Serial Correlation: Bertrand, Duflo, and Mullainathan (2004)

Question. Difference-in-differences studies (Week 10) regress an outcome on a policy dummy plus state and year fixed effects. Are their standard errors trustworthy?
Data. CPS earnings of women aged 25–50, 1979–1999: 50 states $\times$ 21 years.

- **Placebo test**: invent a “law” (a random half of the states, a random onset year) and estimate its effect on wages: *zero by construction.*
- Conventional standard errors called the fake law “significant” at 5% in 45% of the placebo runs.
- Cause: a state’s wage shocks are strongly correlated *across years*, but the SEs treated the 50×21 state-years as independent: the surviving covariances again, now over time.
- Clustering by state (arbitrary correlation across years) brought rejection rates back near 5%.

The rule for policy-adoption designs: cluster at the level of the adopting unit.

Bootstrap

The Idea ★

Everything so far derived a *formula* for the sampling distribution. The bootstrap instead simulates it.

Plug-in principle (Efron 1979)

We want the sampling distribution of a statistic when samples are drawn from the population F. We do not know F, but the sample *is* our best estimate of F. So draw new samples from the sample.

- Especially valuable for statistics with no convenient closed-form variance: quantiles, ratios, decompositions, two-step estimators, indirect effects.
- It also *exposes* fragility that a formula hides. With an influential outlier, replications that happen to draw it look very different from those that miss it: the histogram goes lumpy or skewed, and the interval honestly widens.

The Nonparametric (Pairs) Bootstrap ★

Notation: θ is any target, $\hat{\theta}$ its estimate, B the number of resamples.

1. For $b = 1, \ldots, B$: draw n observations (Y_i, X_i) *with replacement* from the sample.
2. Re-estimate on the resample to get $\hat{\theta}^{(b)}$.
3. Use the empirical distribution of $\{\hat{\theta}^{(b)}\}_{b=1}^{B}$.

- se_{boot} = the standard deviation of the $\hat{\theta}^{(b)}$.
- **Percentile CI**: the 2.5th and 97.5th percentiles of $\{\hat{\theta}^{(b)}\}$.
- Because it resamples (Y_i, X_i) jointly, the pairs bootstrap is automatically heteroskedasticity-robust, given *independent* observations; it assumes nothing about how the error variance moves with X.

Confidence Intervals from the Bootstrap ★

Two ways to turn the $\hat{\theta}^{(b)}$'s into a 95% interval:

- **Normal-approximation**: $\hat{\theta} \pm 1.96\,\mathrm{se}_{\mathrm{boot}}$, with $\mathrm{se}_{\mathrm{boot}}$ the standard deviation of the $\hat{\theta}^{(b)}$. Uses the bootstrap only for the *width*, then assumes the bell shape.
- **Percentile**: the 2.5th and 97.5th percentiles of the $\hat{\theta}^{(b)}$ themselves. Uses the whole shape.

Is the bootstrap distribution normal? Not necessarily. The histogram of the $\hat{\theta}^{(b)}$ estimates the *shape* of the sampling distribution: for large n it usually goes bell-shaped (the same CLT as always), but at your actual n it can be visibly skewed. When it is, the percentile interval follows the skew and the normal-approximation interval misses it. Look at the histogram; when the two intervals disagree, that shape is the reason.

$B = 1{,}000$ is enough for the SE. Percentile endpoints live in the tail, so intervals and p-values want $B \geq 5{,}000$.

Bootstrapping Clustered Data ★

Bootstrap the design: resample the units the lottery ran over.

- Resampling *individuals* from clustered data reproduces the too-small standard error just as faithfully as the naive formula does: the bootstrap copies whatever sampling scheme you feed it.
- **Cluster (block) bootstrap**: draw G whole clusters *with replacement*, each drawn cluster bringing all its people. A pseudo-sample repeats some clusters and omits others: the variation across replications is *between-cluster* variation, as it should be. Resample at the level you would cluster the SEs: treatment at state, resample states; GSS tracts within states, still states.

Limits

Freedman's Critique

Freedman (2006) makes an uncomfortable point.

The argument

The sandwich gives a consistent estimate of the variance of $\hat{\beta}$ *around the population regression coefficient* β. If the model is misspecified, if the CEF is not $X_i'\beta$, then β is merely the best linear approximation, and a correct standard error for a quantity you do not care about is small comfort.

- Distinguish three failures: a **nonlinear CEF** (OLS still consistently estimates the BLP from Week 1, so robust inference about *that* is fine); an **exogeneity failure** ($\hat{\beta}$ no longer estimates the causal quantity you wanted); a **wrong estimand** (the BLP may not answer the question).
- Robust standard errors repair neither of the last two: precise inference about the wrong quantity is small comfort.

A Scoreboard ★

Problem	Robust SEs?	What actually helps
Heteroskedasticity	**yes**	(given independent observations)
Within-cluster correlation	**yes** (cluster-robust)	design-based clustering
Serial correlation in panels	**yes**	cluster on unit
Non-normal errors	**yes** (large-n approx.)	larger n
Omitted variable bias	no	identification strategy (Wks 5–11)
Simultaneity, reverse causation	no	a design (Weeks 7–10)
Measurement error in X	no	IV, validation data
Nonlinear CEF (CEF the target)	no	saturate, splines, flexible ML
Selection into the sample	no	modelling the selection
Wrong estimand	no	say what you are estimating
Influential outliers	no	diagnostics; robust regression
p-hacking, specification search	no	pre-registration

A Practical Protocol ★

1. State the estimand and the design *before* choosing a standard error.
2. **Identify the independently sampled or independently assigned units, and cluster there.** Use HC1 or HC3 only when observations are plausibly independent.
3. If robust and classical standard errors differ dramatically, treat it as a diagnostic: something about the specification or the influence structure deserves attention.

Looking Ahead

- We now have a complete account of OLS: what it estimates, and how to quantify our uncertainty about it.
- **Weeks 3–4**: maximum likelihood. The same asymptotic machinery, score, information, sandwich, returns in a more general form, and OLS turns out to be a special case.
- **Weeks 5+**: the harder question. Not "how precise is $\hat{\beta}$?" but "is β the thing I want?"

Problem Set 1

Assigned today, due Monday September 14 at 11:59PM. Covers Weeks 1–2: scalar OLS algebra, FWL, omitted variable bias, and robust/clustered inference on the GSS extract.

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